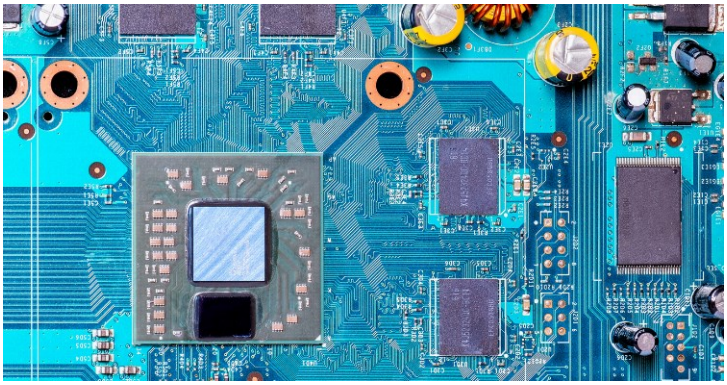
XIAMEN UNIVERSITY MALAYSIA

ASSIGNMENT 1



Lecture 1-7



Dr. Mohammadmahdi Ariannejad

mohammadmahdi.ariannejad@xmu.edu.my

Try to solve the questions **Completely with all steps**! (Effective for grading)

The answer can be in **ONE PDF** format file only (individual activity)

Cheating in any condition will **consider zero Mark** in any exam and assignment!

only **1 Attempt** is allowed.

The deadline is the **24th of May 2023 at 11:00 AM**

Late submission will impose **a penalty (communicate with me if delay happened)**!

Cheating and writing from **Internet resources** will consider as cheating, please avoid any kind of cheating! (**Do not share answers** at any cost!)

The answers will be check for ChatGpt usage; will cause zero marks.

Please use the **Assignment template**, otherwise your script will not be graded.

1. Assume a memory access to main memory on a cache "miss" takes 30 ns and a memory access to the cache on a cache "hit" takes 3 ns. If 80% of the processor's memory requests result in a cache "hit", what is the average memory access time? (explain your answer)

2. The following cache represents a 2-way set associative cache, i.e., there are two lines per set. Notice that the set ID values start at 01101101_2 and increment every other row. This is meant to imply that you are looking at a group of lines/sets toward the middle of the cache and not the entire cache. There are 14 bits for the tag, 8 bits for the set id, and 2 bits for the word id. Answer the following 3 questions based on this cache.

Tag (binary values)	Set ID (binary values)	Word within block			
		00	01	10	11
10100001001001	01101101	00_{16}	61_{16}	$C2_{16}$	23_{16}
11100001100100	01101101	10_{16}	71_{16}	$D2_{16}$	33_{16}
11001011010110	01101110	20_{16}	81_{16}	$E2_{16}$	43_{16}
11100101101011	01101110	30_{16}	91_{16}	$F2_{16}$	53_{16}
11110110110100	01101111	40_{16}	$A1_{16}$	02_{16}	63_{16}
10100111010101	01101111	50_{16}	$B1_{16}$	12_{16}	73_{16}
10101010111110	01110000	84_{16}	$E5_{16}$	46_{16}	$A7_{16}$
10101010010011	01110000	94_{16}	$F5_{16}$	56_{16}	$B7_{16}$
01110001001000	01110001	$A4_{16}$	$A5_{16}$	66_{16}	$C7_{16}$
00001101101101	01110001	$B4_{16}$	15_{16}	76_{16}	$D7_{16}$
01011010010010	01110010	$C4_{16}$	25_{16}	86_{16}	$E7_{16}$
10101111001011	01110010	$D4_{16}$	35_{16}	96_{16}	$F7_{16}$

a) A copy of the data from memory address 7121C5 (hex) is contained in the portion of the cache shown above. Find the value that was retrieved from that address in the space below as a two-digit hexadecimal number with no base identification, e.g., 0x4F (hexadecimal 4F) should be entered as 4F.
b) How many lines are contained in this cache?
c) How many blocks are contained in the memory space (not the cache, but the memory) of the cache system defined above? (Due to D2L constraints, the notation used to represent 2 raised to the power of n is 2^n)

3.

Given a 32 × 8 ROM chip with an enable input, show the external connections necessary to construct a 128 × 8 ROM with four chips and a decoder.

4. The ENIAC was a decimal machine, where a register was represented by a ring of 10 vacuum tubes. At any time, only one vacuum tube was in the ON state, representing one of the 10 digits. Assuming that ENIAC had the capability to have multiple vacuum tubes in the ON and OFF state simultaneously, why is this representation “wasteful” and what range of integer values could we represent using the 10 vacuum tubes?

5. Consider a hypothetical microprocessor generating a 16-bit address (for example, assume that the program counter and the address registers are 16 bits wide) and having a 16-bit data bus.

   a) What is the maximum memory address space that the processor can access directly if it is connected to a "16-bit memory"?

   b) What is the maximum memory address space that the processor can access directly if it is connected to an "8-bit memory"?

   c) What architectural features will allow this microprocessor to access a separate "I/O space"?

   d) If an input and an output instruction can specify an 8-bit I/O port number, how many 8-bit I/O ports can the microprocessor support? How many 16-bit I/O ports? Explain.

6. Consider a computer system that contains an I/O module controlling a simple keyboard/printer teletype. The following registers are contained in the processor and connected directly to the system bus:

INPR:	Input Register, 8 bits
OUTR:	Output Register, 8 bits
FGI:	Input Flag, 1 bit
FGO:	Output Flag, 1 bit
IEN:	Interrupt Enable, 1 bit

Keystroke input from the teletype and printer output to the teletype are controlled by the I/O module. The teletype is able to encode an alphanumeric symbol to an 8-bit word and decode an 8-bit word into an alphanumeric symbol.

a) Describe how the processor, using the first four registers listed in this problem, can achieve I/O with the teletype.

b) Describe how the function can be performed more efficiently by also employing IEN.

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